

Supplementary Materials

Residuals from a Load-Normalized Spine-Neck Divider Reveal Limits of Scalar Isolation Metrics

1 Extended Equations and Unit Notes

The implementation uses SI units internally at solver boundaries. Voltages are converted from mV to V, capacitances from pF to F, conductances from nS to S, currents from pA to A, and time from ms to s. Geometry-derived neck resistance uses centimeters for ρ_i compatibility:

$$R_{\text{neck}} = \frac{\rho_i L_n}{\pi r_n^2}. \quad (1)$$

For nonuniform neck profiles the code evaluates

$$R_{\text{neck}} = \int_0^{L_n} \frac{\rho_i}{\pi r(x)^2} dx \quad (2)$$

by numerical quadrature. Tapered, constricted, and profile-based necks preserve the intended geometry and report an effective resistance for comparison with lumped approximations.

The spine-omitted dendrite-soma load is a two-node DC circuit in the baseline reference case. With dendrite leak $g_{L,d}$, soma leak $g_{L,s}$, and dendrite-soma coupling g_{ds} , direct linear solution gives the dendritic one-port input resistance used in the main text:

$$R_{\text{in},d} = \left(g_{L,d} + \frac{g_{ds} g_{L,s}}{g_{ds} + g_{L,s}} \right)^{-1}. \quad (3)$$

Using the baseline values gives $R_{\text{in},d} = 144.486692 \text{ M}\Omega$, matching the independent direct DC solve to round-off.

The analytic local divider is therefore

$$\rho_L = \frac{R_{\text{neck}}}{R_{\text{in},d}}, \quad \Gamma_{\text{div}} = \frac{R_{\text{in},d}}{R_{\text{neck}} + R_{\text{in},d}} = \frac{1}{1 + \rho_L}, \quad (4)$$

and the manuscript residual is

$$\Delta_{\text{div}} = \Gamma_{h \rightarrow d} - \Gamma_{\text{div}}. \quad (5)$$

These equations define a DC small-signal reference. SMI is retained in code-facing outputs as legacy shorthand for ρ_L . The monotone relation between ρ_L and Γ_{div} is analytic by construction and is not an empirical discovery. These equations do not assert that transient conductance-based peak transfer must equal the divider value.

The passive matrix in the main manuscript generalizes to an arbitrary passive tree as

$$\mathbf{C} \frac{d\mathbf{V}}{dt} = -(\mathbf{G}_L + \mathbf{G}_A + \mathbf{G}_{\text{syn}}(t)) \mathbf{V} + \mathbf{b}(t), \quad (6)$$

where $\mathbf{G}_A$ is assembled from axial edges. Each edge with conductance g_{ij} contributes $+g_{ij}$ to both diagonal entries and $-g_{ij}$ to the symmetric off-diagonal entries. This construction preserves reciprocity in passive linear systems.

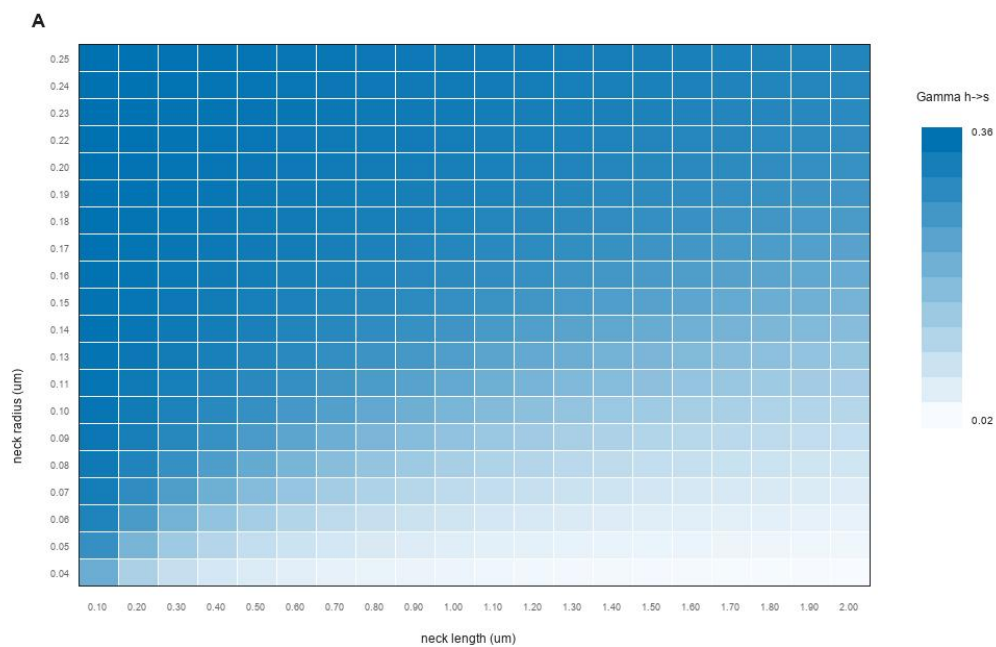


Figure S1. Fixed-load head-to-soma transfer heatmap. Baseline validation sweep complementing the main-text load-normalization controls by showing the two-parameter neck-geometry grid.

2 Numerical Verification and Benchmarks

The passive baseline was checked against geometry identities, synaptic normalization, current signs, matrix limits, timestep refinement, high-synapse stability, and Backward Euler versus Crank-Nicolson agreement. The baseline verification script generated the passive source data and classified all representative reference target values as passing workflow-specified target-case tolerances. The most stringent target-case tolerance class means absolute error ≤ 0.005 or relative error $\leq 2\%$, not bitwise equality.

An independent direct-matrix benchmark was added for the passive three-compartment baseline reference cases. The benchmark independently parsed the baseline parameters, recomputed capacitance/leak/neck terms and the double-exponential synaptic conductance, assembled the passive matrices, and solved the Backward Euler system without calling the production passive simulation function as its benchmark engine. Compared with existing SPINE trace CSVs, the maximum all-trace absolute difference was 1.25×10^{-13} mV, the maximum head-amplitude difference was 1.42×10^{-14} mV, and the maximum $\Gamma_{h \rightarrow d}$ difference was 7.55×10^{-15} .

DC analytic limits were also checked. The spine-omitted two-node load formula gave $R_{in,d} = 144.48669201520912$ M Ω , and the direct DC solve gave 144.48669201520914 M Ω . Divider predictions for the low, intermediate, and high reference cases were 0.9912648, 0.8970688, and 0.4306959. The zero-neck and very-large-neck limiting cases approached local transfer values of 1 and 0, respectively.

The BE-versus-CN reference-case comparison remains an internal scheme-consistency check. Across the three baseline reference cases, the maximum peak differences were 0.00203 mV for A_h , 0.000534 mV for A_d , 0.000189 mV for A_s , 0.000500 for $\Gamma_{h \rightarrow d}$, and 0.000180 for $\Gamma_{h \rightarrow s}$. The maximum trace voltage difference was 0.112 mV in the high- ρ_L case.

Passive morphology verification included sparse versus dense equivalence, spatial convergence, sinusoidal impedance checks, logarithmic-chirp impedance checks, and distributed-neck comparisons. Sinusoidal checks produced maximum relative amplitude error 0.0054. Chirp checks produced maximum relative amplitude error 0.072. Spatial convergence passed at the 16-versus-32 segment comparison with relative $\Gamma_{h \rightarrow s}$ difference 0.014.

Active-mechanism verification included AMPA peak normalization, NMDA magnesium-block monotonicity, NMDA current-voltage relief, active-current sign checks, gate bounds, positive gate time constants, resting-state stability, timestep refinement, limiting-case solver comparison, strong-drive stability, and active unit checks. All 12 active verification rows passed.

NEURON was not importable in the project runtime and no NEURON result is claimed. Brian2 was installed only in a project-local optional validation environment and used for Brian2-hosted scoped checks of the three passive baseline cases, one cylindrical distributed-neck passive case, and one AMPA/NMDA synapse-only case. These checks used Brian2 as a unit, clock, and state host with an independent SI backward-Euler update. They passed predefined tolerances and provide representative implementation-check support, but they do not validate all rows, restrained active channels, or biological calibration. Brian2 is not a required SPINE runtime dependency.

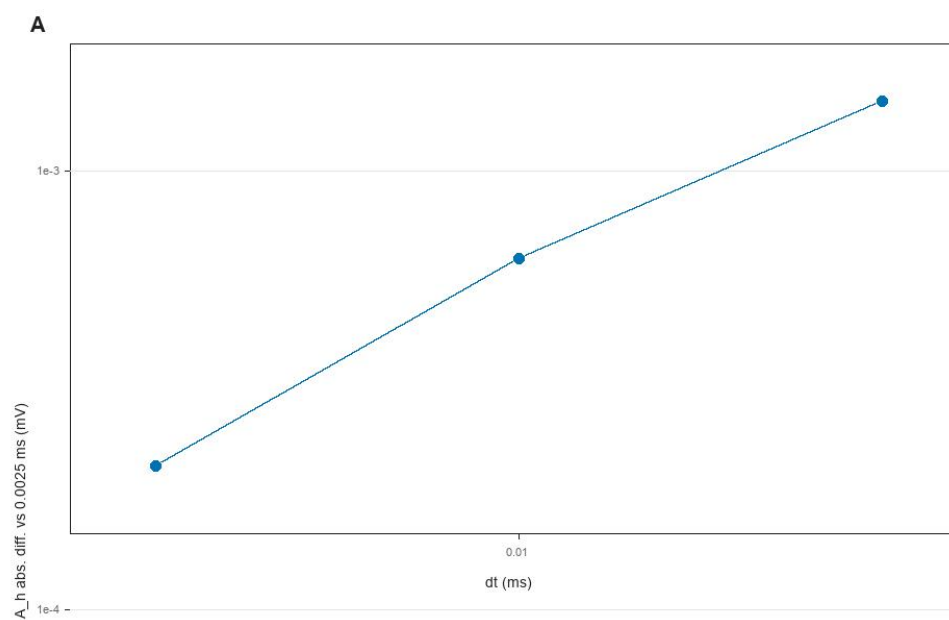


Figure S2. Timestep convergence. Baseline-reference intermediate condition showing head-amplitude absolute difference relative to the finest audited timestep on log-scaled axes.

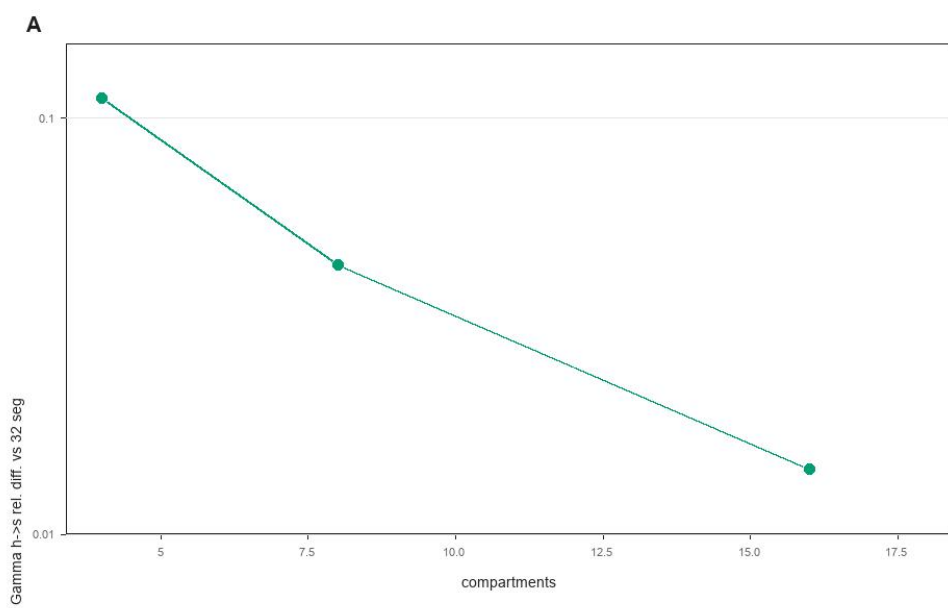


Figure S3. Spatial convergence. Passive morphology-engine convergence shown as relative head-to-soma transfer difference against the 32-segment reference.

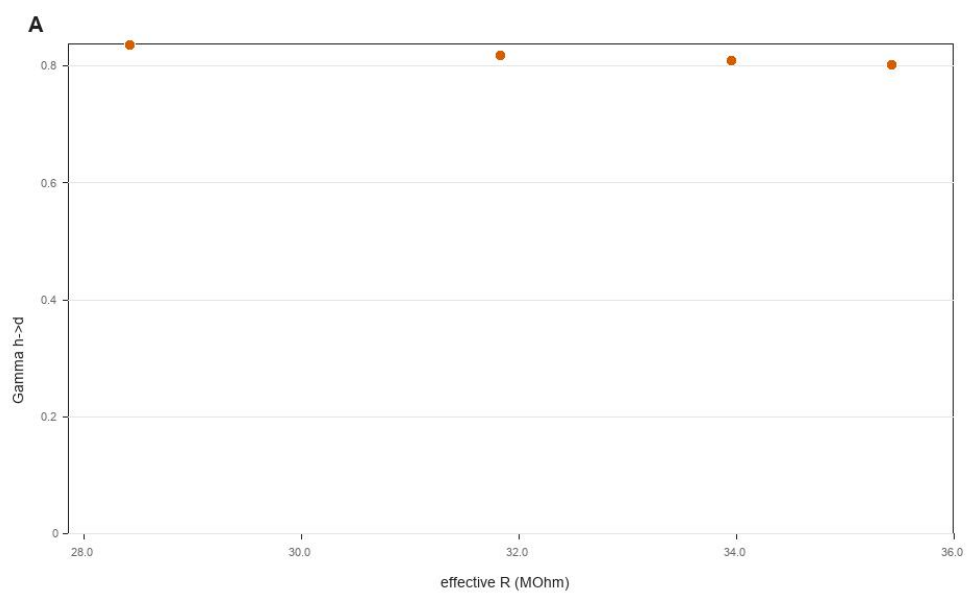


Figure S4. Lumped and distributed neck-model comparisons. Effective neck resistance contrasted with local transfer for audited passive neck variants.

3 Release, Environment, and Residual-Control Provenance

Residual timing controls, row provenance, validation coverage, high- ρ_L stress-test summaries, and environment metadata are reported once in the tables at the end of this supplement. The residual-control table reports the manuscript’s separate-peak metric alongside matched-head-peak and area-under-curve controls, because the DC divider is a low-frequency reference and the simulated traces are transient conductance-driven responses.

The reconciled row-provenance table is the authoritative denominator accounting. It reports 3,718 included deterministic design-derived residual rows, with 3,584 non-exploratory rows and 134 exploratory stress-test rows. The original N=96 uncertainty screen is retained only as archived provenance when listed separately; it has denominator contribution 0 in that listing because adding it again would create the erroneous 3,910-row total.

The environment table separates the default project/document runtime from the optional Brian2 validation environment. Package availability therefore should be read by context, not as a statement that every dependency was importable in every runtime or required by the SPINE solver.

4 Uncertainty, Identifiability, and Predictor Stability

The uncertainty analysis used deterministic Latin-hypercube ensembles and deterministic resampling diagnostics. A robustness audit extended this to progressive N=96, 192, 384, and 768 ensembles. The global uncertainty seed was 202605; progressive-sampling ensembles used deterministic offsets $202605 + 5100 + N$. The original 48-versus-96 ρ_L convergence failure was preserved and interpreted as a small-sample warning in a skewed ρ_L distribution. By N=768, final ρ_L median convergence passed with relative change 0.012, and final all-median convergence passed with maximum relative change 0.029.

Radius uncertainty was analyzed independently because of the quadratic radius term in R_{neck} . The radius audit used 25 deterministic offsets spanning ± 2 SD, with $\text{SD} = 0.018 \mu\text{m}$ and clipping to 0.06–0.22 μm . At N=768, 184 of 768 sampled parameter combinations changed ρ_L class under the analysis-defined radius-error model, a design fraction of 24.0%. Low-class combinations flipped in 22.9% of cases, all 11 intermediate combinations flipped, and no high- ρ_L combinations were present. The median radius-induced ρ_L fold range was 3.44. These values are deterministic design summaries, not biological rates.

High- ρ_L uncertainty coverage is absent in the main N=768 deterministic ensemble. The ensemble contains 757 low- ρ_L combinations, 11 intermediate combinations, and zero high- ρ_L combinations. High-ratio behavior in the manuscript is therefore supported by reference, designed challenge cases, and targeted extension rows rather than by the N=768 uncertainty ensemble.

Predictor comparisons used Pearson association, Spearman association, deterministic resampling summaries, cross-validated RMSE, and multivariable diagnostics. Exact top predictor labels were not fully stable because near-tied dynamic-ratio variants changed order. The manuscript therefore emphasizes descriptor families and mechanistic interpretation.

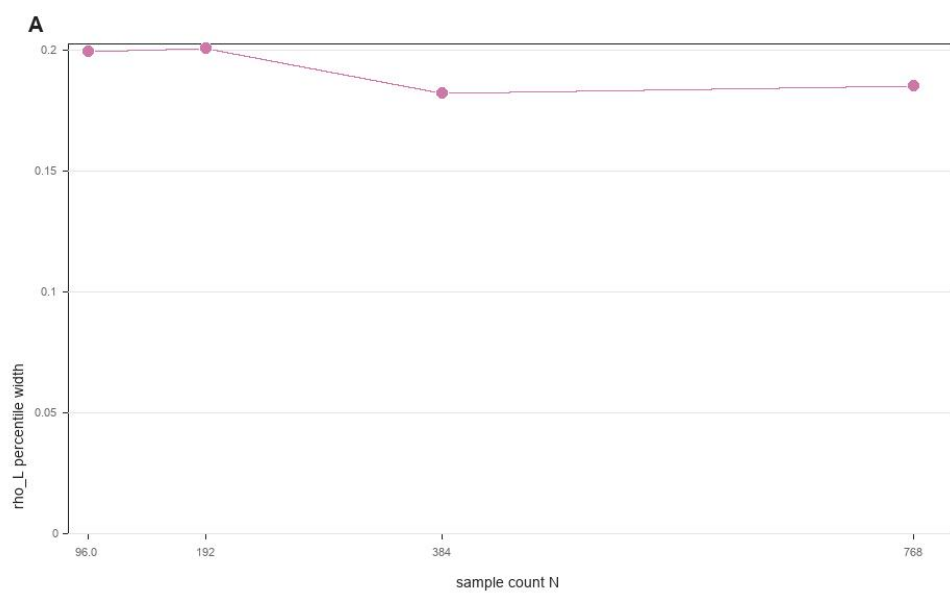


Figure S5. Continuous ρ_L uncertainty-width convergence. Progressive sample counts for the continuous-width check complement the main-text radius class-flip design fraction.

5 Active and Nonlinear Mechanism Notes

AMPA and NMDA synapses are opt-in extensions. NMDA includes a voltage-dependent magnesium block based on established reduced forms [1, 2]. Na and delayed-rectifier K conductances follow Hodgkin-Huxley-style conductance-current logic [3]. HCN, A-type K, and restrained calcium conductances are included as generic active-state stressors with provenance from dendritic physiology literature [4–7].

These conductances are not cell-type-calibrated densities. They are disabled in the baseline reference configuration and should not be interpreted as fitted biological parameters. Their purpose in this manuscript is to test whether passive load-ratio conclusions survive nonlinear and active stress tests. The implementation map below links each mechanism to its parameterization; source-code functions are listed in the accompanying reproducibility reports and public release manifest.

Table S1. Active-extension parameter audit. All mechanisms are disabled in the baseline reference configuration and enabled only through active-extension configurations or explicit protocols.

Mechanism	Main parameters	Values	Units	Provenance/status
AMPA	$g_{\max}$; τ_r ; τ_d ; E_{rev}	1.4; 0.3; 3.0; 0	nS; ms; ms; mV	active-extension double exponential
NMDA	$g_{\max}$; τ_r ; τ_d ; Mg ; η ; γ	0.7; 2.0; 80.0; 1.0; 3.57; 0.062	nS; ms; ms; mM; mM; mV^{-1}	reduced magnesium block family
Na	$\bar{g}$; E_{rev} ; gates	6.0; 55; m^3h	nS; mV; dimensionless	restrained HH-style extension
KDR	$\bar{g}$; E_{rev} ; gates	1.5; -90; n^4	nS; mV; dimensionless	restrained HH-style extension
HCN	$\bar{g}$; E_{rev} ; gates	0.05; -30; r	nS; mV; dimensionless	generic subthreshold HCN approximation
A-type K	$\bar{g}$; E_{rev} ; gates	0.5; -90; a^3b	nS; mV; dimensionless	restrained transient potassium extension
CaT	$\bar{g}$; E_{rev} ; gates	0.05; 120; p^2q	nS; mV; dimensionless	electrical-only restrained calcium extension

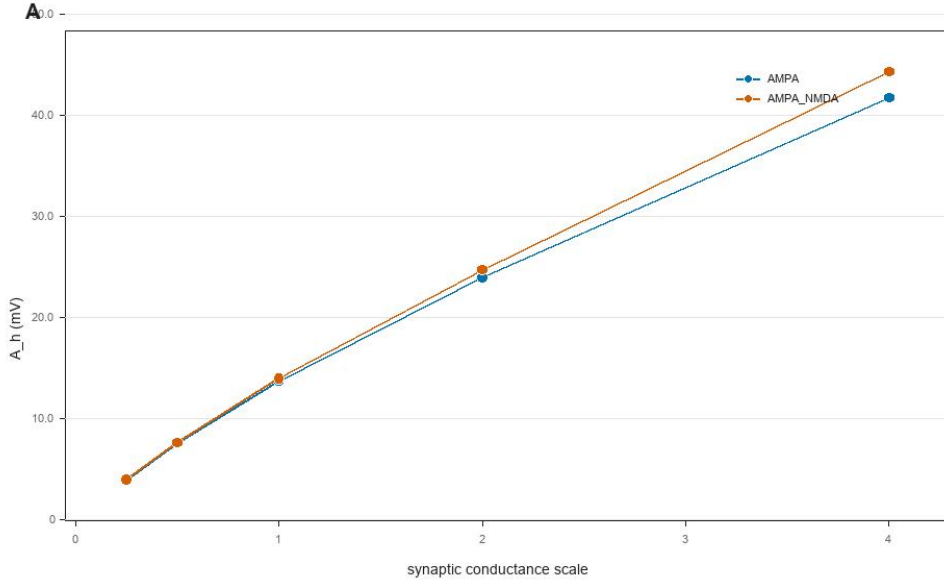


Figure S6. Active AMPA and AMPA+NMDA strength-sweep protocol. Head-amplitude responses across analysis-defined conductance scales for the active-extension protocol library.

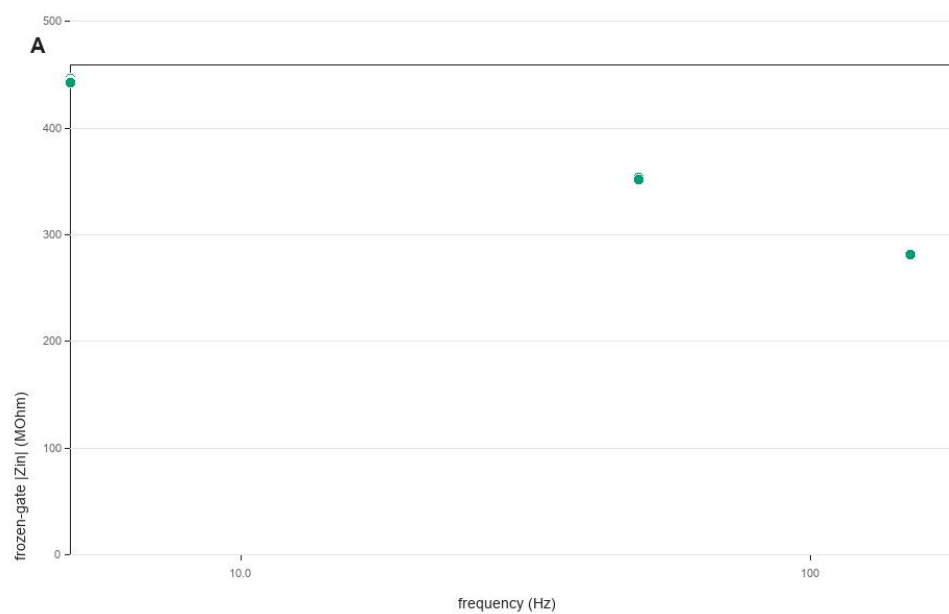


Figure S7. Exploratory frozen-gate operating-point impedance. Operating-point impedance estimates used as exploratory active-context descriptors.

6 Exploratory Perturbation Scenarios

The separated exploratory module uses restrained perturbation scenarios involving morphology, HCN, potassium, AMPA/NMDA, clustering, synchrony, combined perturbations, and conflicting directions [8–12]. The source rows explicitly mark these scenarios as exploratory-only and make no diagnostic or therapeutic claim.

The exploratory uncertainty screen uses $N=8$ sampled combinations per scenario. It is a sensitivity screen rather than a biological uncertainty estimate or rare-tail analysis. The exploratory scenarios are best cited as examples showing that synaptic strength, active state, clustering, and timing can alter voltage outcomes without changing morphology-defined ρ_L .

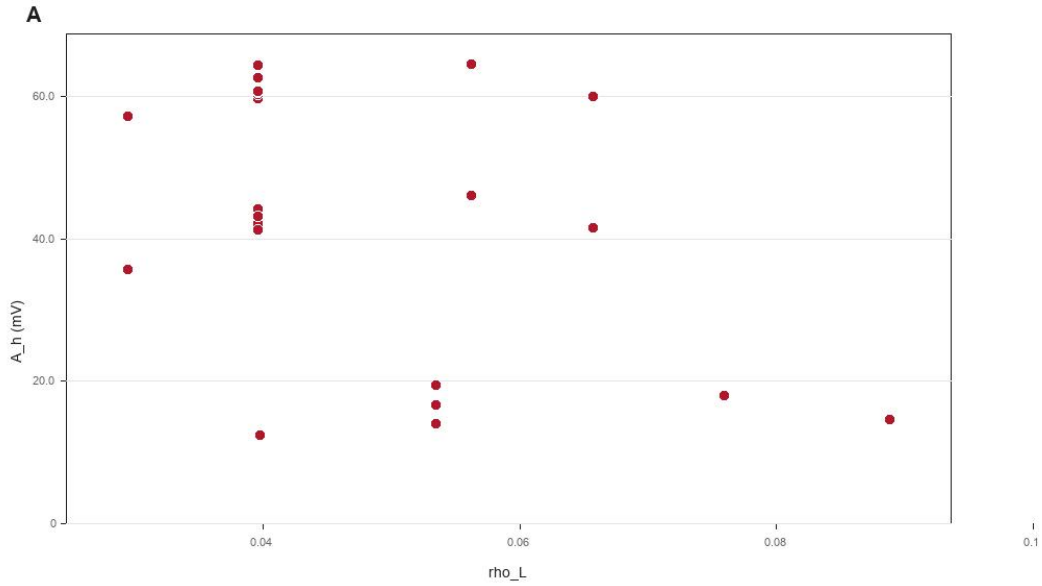


Figure S8. Exploratory perturbation scenarios. ρ_L versus head amplitude for cases separated from the general model and not used to support primary claims.

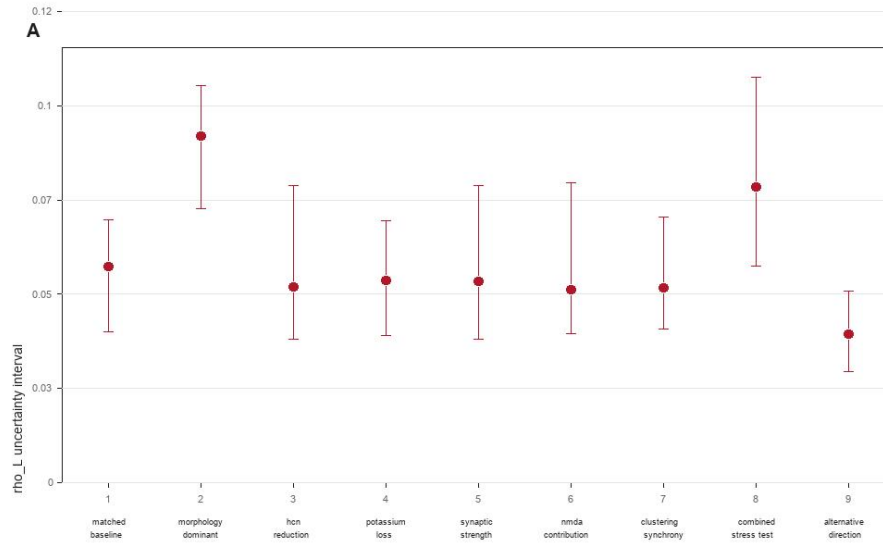


Figure S9. Scenario-level ρ_L uncertainty screen. Intervals summarize exploratory perturbation scenarios and are presented as context rather than primary validation evidence.

7 Reproducibility Commands

The following commands reproduce the validated software checks and manuscript source-data generation in a local development environment. Broad primary ensembles should not be rerun casually during review unless the authors intentionally regenerate the source data.

Windows PowerShell:

```
$env:PYTHONPATH='src'
python -m unittest discover -s tests -p 'test_*.py'
python -m spine.cli smoke configs\manuscript_faithful\baseline.toml '
--output-csv results\phase01_smoke_trace.csv
python scripts\reproduce_manuscript.py
python scripts\revision_restart\phase4_validation_runner.py
python scripts\revision_restart\phase7_generate_divider_residual_figure.py
```

Unix shell:

```
PYTHONPATH=src python -m unittest discover -s tests -p 'test_*.py'
PYTHONPATH=src python -m spine.cli smoke configs/manuscript_faithful/baseline.toml \
--output-csv results/phase01_smoke_trace.csv
PYTHONPATH=src python scripts/reproduce_manuscript.py
PYTHONPATH=src python scripts/revision_restart/phase4_validation_runner.py
PYTHONPATH=src python scripts/revision_restart/phase7_generate_divider_residual_figure.py
```

The public submission release is available at <https://github.com/blakepi/spine-divider-residuals>, tagged as v1.0.0-submission, and archived at <https://doi.org/10.5281/zenodo.20672333>. Code is released under the MIT License; manuscript-facing derived data, figures, tables, and documentation are released under CC BY 4.0 unless otherwise stated.

8 External Summary Anchor Decision Gate

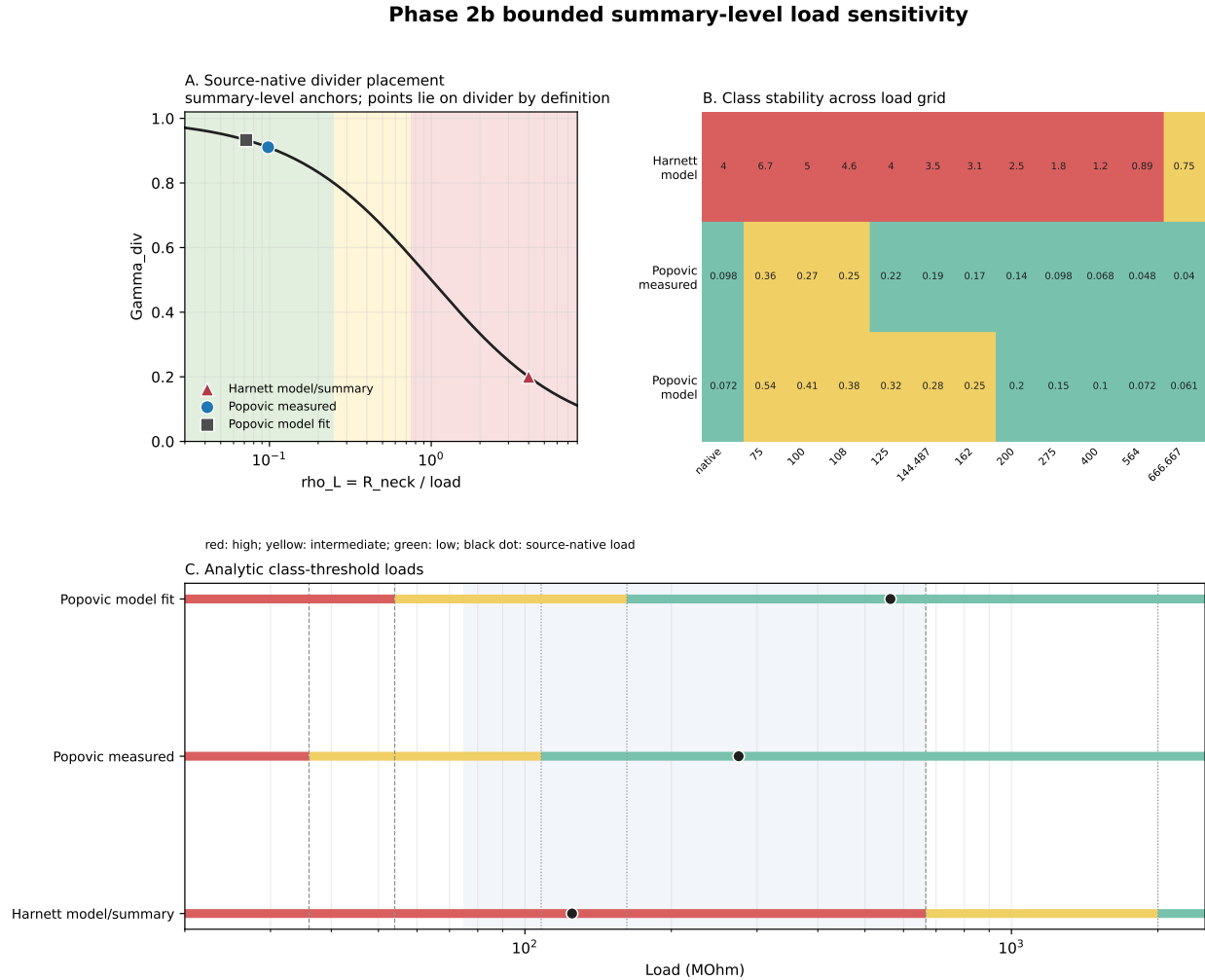


Figure S10. Bounded summary-level load sensitivity for external Route 1 anchors.

Source-native summary anchors are plotted on the analytic divider relationship $\Gamma_{\text{div}} = 1/(1 + \rho_L)$, and points lie on the divider by definition rather than as residual validation. The Harnett point is a model/summary anchor, not recovered paired measured per-spine data. Popovic measured summary and Popovic model-fit rows are kept separate. Class stability is shown across source-native, fixed/recovered common-load, and deterministic diagnostic-load calculations; the diagnostic grid is not a biological distribution. Threshold bars show the loads required for each row to cross low/intermediate/high ρ_L classes. These rows support only a summary-level load-normalization hypothesis; no per-spine inference, biological validation, prevalence estimate, controversy resolution, or SPINE residual claim from transfer proxies is implied.

9 Extended Limitations

SPINE is a transparent computational framework, not a calibrated biological reconstruction. The baseline reference track preserves pre-specified assumptions even when they are simplified. The plausibility-revised and active-extension tracks are explicitly separated. Exploratory perturbation scenarios are supplemental stress tests and are not disease models.

The analytic divider is a DC small-signal reference. The simulated targets in this manuscript are transient conductance-based peak responses, so residuals from the divider are expected and informative. The independent direct-matrix benchmark checks the passive three-compartment baseline implementation against an independently assembled equation solve, but it does not check every morphology, active, uncertainty, or exploratory output. NEURON cross-checking was not performed.

The deterministic uncertainty ensembles are sensitivity designs, not experimentally fitted posterior distributions. The $N=768$ ensemble contains no high- ρ_L rows, so high-isolation uncertainty remains a targeted limitation. Radius uncertainty, unmodeled ultrastructure, cytoplasmic resistivity uncertainty, segmentation error, local synaptic receptor composition, and active conductance distributions can all alter interpretation.

The current package does not include posterior inference from experimental voltage traces, cell-type-specific channel fitting, full morphology libraries, biochemical calcium reaction-diffusion, receptor trafficking, signaling cascades, or receptor-level plasticity mechanisms. These are future validation targets, not hidden assumptions of the current manuscript.

References

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Table S2. Validation coverage by deterministic row class. Brian2-hosted checks are scoped and do not cover every row.

row_class	row_count	direct_matrix_benchmark_coverage	dc_analytic_ch
baseline targets	3	yes	yes
fixed-load sweep	400	partial	yes
matched-neck/matched-load sweep	30	partial	yes
passive morphology	24	no	partial
spatial convergence/distributed morphology checks	4	partial	partial
active/NMDA challenge	47	no	no
progressive uncertainty designs	2880	no	ratio algebra on
original N=96 uncertainty screen	192	no	ratio algebra on
exploratory perturbation scenarios	134	no	ratio algebra on

Table S3. Reconciled deterministic row provenance. The archived original N=96 uncertainty screen is non-additive.

row_class	row_count	denominator_contribution	status	ex
phase02_baseline_target_cases	3	3	included	n
phase02_dt_convergence_intermediate	4	4	included	n
phase02_fixed_load_geometry_sweep	400	400	included	n
phase02_matched_neck_load_sweep	30	30	included	n
phase03_passive_morphology_challenge	24	24	included	n
phase03_spatial_convergence	4	4	included	n
phase04_active_nonlinear_challenge	47	47	included	n
phase05_1_global_uncertainty_N96	192	192	included	n
phase05_1_global_uncertainty_N192	384	384	included	n
phase05_1_global_uncertainty_N384	768	768	included	n
phase05_1_global_uncertainty_N768	1536	1536	included	n
phase05_global_uncertainty_N96	192	192	included	n
phase06_exploratory_matched_baseline_comparisons	27	27	included	ye
phase06_exploratory_mechanistic_decomposition	8	8	included	ye
phase06_exploratory_scenario_metrics	27	27	included	ye
phase06_exploratory_uncertainty_samples	72	72	included	ye
original_N96_uncertainty_screen_if_listed_separately	192	0	excluded/nested	n

Table S4. Residual metric controls for baseline cases and the largest matched-neck peak residual.

condition	rho_load	Gamma_divider	Gamma_peak_separate_h_to_d
low	0.0088121579017196	0.9912648179022263	0.9839703151549645
intermediate	0.1147416393453083	0.8970688495921831	0.751282130869046
high	1.3218236852579517	0.4306959250822274	0.10213244622161506
matched_neck_largest_peak_residual	0.45210526315789473	0.6887	0.1962

Table S5. Row-class residual summaries. Fractions are deterministic design fractions.

row_class	n	median_abs_residual	mean_abs_residual	max_abs_residual
phase02_baseline_target_cases	3	0.1458	0.1605	0.3286
phase02_dt_convergence_intermediate	4	0.1457	0.1458	0.1463
phase02_fixed_load_geometry_sweep	400	0.1258	0.1779	0.4201
phase02_matched_neck_load_sweep	30	0.4313	0.378	0.4924
phase03_passive_morphology_challenge	24	0.1071	0.1339	0.454
phase03_spatial_convergence	4	0.02423	0.02465	0.02675
phase04_active_nonlinear_challenge	47	0.09292	0.1039	0.1938
phase05_1_global_uncertainty_N192	384	0.05347	0.05815	0.17
phase05_1_global_uncertainty_N384	768	0.05246	0.05704	0.1437
phase05_1_global_uncertainty_N768	1536	0.05315	0.05739	0.1724
phase05_1_global_uncertainty_N96	192	0.05391	0.05651	0.1142
phase05_global_uncertainty_N96	192	0.05299	0.05753	0.1362
phase06_exploratory_matched_baseline_comparisons	27	0.02555	0.02976	0.05912
phase06_exploratory_mechanistic_decomposition	8	0.0388	0.04189	0.05212
phase06_exploratory_scenario_metrics	27	0.02555	0.02976	0.05912
phase06_exploratory_uncertainty_samples	72	0.04043	0.04169	0.06926

Table S6. Bounded deterministic high- ρ_L uncertainty extension.

metric	value	interpretation
true_high_rhoL_parameter_combinations	41	bounded deterministic high-ratio extension; not a population sample
radius_measurement_rows	205	five deterministic radius scales per true high-ratio combination
class_flip_rows	16	flips occur when measured radius moves borderline high cases below
class_flip_design_fraction	0.07805	design fraction only
median_peak_residual_high_rhoL	-0.3245	same transient peak metric as main residual table

Table S7. Environment and runtime contexts. Brian2 is optional and used only for scoped Brian2-hosted implementation checks. Package availability is runtime-specific.

Context	OS/platform	Python	Numpy	SciPy	Pandas	Matplotlib	scikit-learn/shogun	Brannl	SciPy/NumPy	Notes
Data set project / document random	Windows-10-10.0.17134-SP4	3.7.0	1.16.0	not applicable	1.0.1	not applicable	not applicable / not installed	not applicable	not applicable	Missing optional context package in this bounded runtime do not imply missing package in other project environments.
Local LTPS document build tests	Windows-10-10.0.17134-SP4	not applicable	not applicable	not applicable	not applicable	not applicable	not applicable	not applicable	not applicable	libSVM is unavailable for the provided hardware (CPU) and (OS) QA for most cases.
OpenAI Brain validation environment	Windows-10-10.0.17134-SP4	3.7.0	1.16.0	1.2.1	1.0.1	1.0.1	not installed	not applicable	not applicable	libSVM is not a required 32-bit runtime dependency (check several selected package and system-only reference cases).
Reference numerical settings	not applicable	not applicable	not applicable	not applicable	not applicable	not applicable	not applicable	not applicable	not applicable	libSVM is not a required 32-bit runtime dependency (check several selected package and system-only reference cases).
Reference metadata	not applicable	not applicable	not applicable	not applicable	not applicable	not applicable	not applicable	not applicable	not applicable	libSVM is not a required 32-bit runtime dependency (check several selected package and system-only reference cases).

Table S8. Active-mechanism implementation map. Active mechanisms are verification stress tests, not biological calibration.

mechanism	source_file	parameters	gating
AMPA	src/spine/synapses.py	gmax 1.4 nS baseline; tau 0.3/3 ms	none
NMDA	src/spine/synapses.py	gmax 0.7 nS active extension; tau 2/80 ms	voltage-dependent Mg unblock
magnesium block	src/spine/synapses.py	Mg 1 mM, eta 3.57, gamma 0.062/mV	voltage-dependent
Na	src/spine/channels.py	na_gbar_nS 6.0	m/h gates
KDR	src/spine/channels.py	kdr_gbar_nS 1.5	n gate
HCN	src/spine/channels.py	hcn_gbar_nS 0.05	h gate
A-type K	src/spine/channels.py	ka_gbar_nS 0.5	a/b gates
restrained CaT	src/spine/channels.py	cat_gbar_nS 0.05	m/h gates